CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · SHORT NOTES

Chapter 20: Cylinder, Cone and Sphere (Surface Area and Volume)

Topic-wise short notes with solved examples

This chapter covers the surface area and volume of the three round solids, hollow solids, solids joined together, and problems in which one solid is melted and recast into another.

1 Cylinder

- A right circular cylinder has radius r and height h.

$$\text{Curved surface area} = 2\pi rh \qquad \text{Total surface area} = 2\pi r(r+h) \qquad \text{Volume} = \pi r^2 h$$

- **Hollow cylinder** (a pipe) with external radius R, internal radius r and height h:

$$\text{Volume of material} = \pi(R^2 - r^2)h \qquad \text{Total surface area} = 2\pi(R+r)h + 2\pi(R^2 - r^2)$$

Example 1.1 A cylinder has radius 7 cm and height 10 cm. Find its curved surface area, total surface area and volume.

SOLUTION

$\text{CSA} = 2 \times \frac{22}{7} \times 7 \times 10 = 440$ cm²

$\text{TSA} = 2 \times \frac{22}{7} \times 7 \times (7+10) = 748$ cm²

$\text{Volume} = \frac{22}{7} \times 7 \times 7 \times 10 = 1540$ cm³

CSA 440 cm²; TSA 748 cm²; volume 1540 cm³

Example 1.2 A metal pipe is 21 cm long, with external radius 4 cm and internal radius 3 cm. Find the volume of metal in it.

SOLUTION

$\text{Volume} = \pi(R^2 - r^2)h = \frac{22}{7} \times (16 - 9) \times 21$

Volume $= 462$ cm³

2 Cone

- A right circular cone has radius r, vertical height h and slant height l.

$$l = \sqrt{r^2 + h^2} \qquad \text{Curved surface area} = \pi rl \qquad \text{Total surface area} = \pi r(l+r) \qquad \text{Volume} = \frac{1}{3}\pi r^2 h$$

- The volume of a cone is one third of the volume of a cylinder with the same base and height.

Example 2.1 A cone has radius 7 cm and height 24 cm. Find its slant height, curved surface area and volume.

SOLUTION

$l = \sqrt{7^2 + 24^2} = \sqrt{625} = 25$ cm

CSA $= \frac{22}{7} \times 7 \times 25 = 550$ cm²

Volume $= \frac{1}{3} \times \frac{22}{7} \times 7 \times 7 \times 24 = 1232$ cm³

$l = 25$ cm; CSA 550 cm²; volume 1232 cm³

3 Sphere and hemisphere

Solid	Surface area	Volume
Sphere	$4\pi r^2$	$\frac{4}{3}\pi r^3$
Hemisphere	Curved: $2\pi r^2$; total: $3\pi r^2$	$\frac{2}{3}\pi r^3$
Spherical shell (radii R, r)	Outer: $4\pi R^2$	$\frac{4}{3}\pi(R^3 - r^3)$

Example 3.1 Find the surface area and the volume of a sphere of radius 10.5 cm.

SOLUTION

Surface area $= 4 \times \frac{22}{7} \times 10.5 \times 10.5 = 1386$ cm²

Volume $= \frac{4}{3} \times \frac{22}{7} \times 10.5 \times 10.5 \times 10.5 = 4851$ cm³

Surface area 1386 cm²; volume 4851 cm³

4 Melting and recasting

- When a solid is melted and recast, or reshaped, its **volume does not change**.

$$\text{Number of small solids} = \frac{\text{volume of the large solid}}{\text{volume of one small solid}}$$

- When a solid is fully immersed in water, the volume of water displaced equals the volume of the solid.

Example 4.1 A solid metal sphere of radius 6 cm is melted and recast into small cones, each of radius 2 cm and height 3 cm. Find the number of cones formed.

SOLUTION

Volume of sphere $= \frac{4}{3}\pi \times 6^3 = 288\pi$ cm³

Volume of one cone $= \frac{1}{3}\pi \times 2^2 \times 3 = 4\pi$ cm³

Number of cones $= \frac{288\pi}{4\pi} = 72$

Example 4.2 A solid metal sphere of radius 9 cm is drawn into a wire of radius 0.2 cm. Find the length of the wire.

SOLUTION

Volume of sphere $= \frac{4}{3}\pi \times 9^3 = 972\pi$ cm³

The wire is a cylinder: $\pi \times (0.2)^2 \times h = 972\pi$, so $h = \frac{972}{0.04} = 24300$ cm

Length of the wire $= 243$ m

Example 4.3 A sphere of radius 3 cm is dropped into a cylindrical jar of radius 6 cm containing water, and is completely submerged. Find the rise in the water level.

SOLUTION

Volume of sphere $= \frac{4}{3}\pi \times 27 = 36\pi$ cm³

Rise h: $\pi \times 6^2 \times h = 36\pi$

The water level rises by 1 cm.

5 Combination of solids

- **Volume**: add the volumes of the parts.
- **Surface area**: add only the surfaces that are exposed. Surfaces along which the solids are joined are not counted.

Example 5.1 A solid is a cone standing on a hemisphere, both of radius 7 cm. The total height of the solid is 31 cm. Find its volume and its surface area.

SOLUTION

Height of cone $= 31 - 7 = 24$ cm, so $l = 25$ cm

Volume $= \frac{1}{3}\pi r^2 h + \frac{2}{3}\pi r^3 = 1232 + \frac{2156}{3} \approx 1950.67$ cm³

Surface area $= \pi r l + 2\pi r^2 = 550 + 308$

Volume ≈ 1950.67 **cm³; surface area** $= 858$ **cm²**

6 Common mistakes

- Using the height h instead of the slant height l in πrl.
- Using the diameter in place of the radius.
- Forgetting that units must match: change mm to cm, or cm to m, before calculating.
- Giving the total surface area of a hemisphere as $2\pi r^2$. A solid hemisphere has $3\pi r^2$.
- Adding surface areas when a problem is about recasting. Recasting keeps the **volume** fixed.

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